Chedy Korbi

AI/ML Engineering Student | LLMs, RAG, Computer Vision, Deep Learning
Tunis | +216 58 239 045 | chedykorbi2021@gmail.com | linkedin.com/in/chedy-korbi | github.com/ChedyKorbi

Profile

AI/ML engineering student (Big Data & AI) building machine learning systems end to end: data pipelines, model training, honest evaluation and deployment behind APIs. Hands-on with LLMs, RAG, computer vision and deep learning, backed by strong software engineering. Seeking AI/ML, LLM or computer-vision roles.

Experience

AI & Mobile Engineering Intern Jul 2026 – Aug 2026
SPIRAW by Food4Future *Tunis, Tunisia*

- Studied whether 23,040 biomass photos can replace lab assays for C-phycocyanin, using leak-free batch splits
- Trained MobileNetV2 in PyTorch: R^2 0.96 on test, 0.87 leave-one-batch-out, beating CatBoost and XGBoost
- Added SHAP and Grad-CAM explainability, optical stress tests and a Flask demo for a future optical sensor
- Rebuilt the EERA assistant with hybrid intent routing (rules, fuzzy matching, LLM fallback) and guardrails that never invent sensor or price data, raising routing accuracy from about 75% to about 100%
- Hardened the Flutter app: authenticated backend in place of direct MQTT, plus a Stripe shop

Final-Year Intern, IoT & AI/ML Jan 2025 – Jun 2025
Symatique Software *Tunis, Tunisia*

- Built breed- and age-calibrated scikit-learn models that flag pet health risks from wearable sensor data
- Streamed ESP32 telemetry over MQTT into inference services for 50+ concurrent users at sub-second latency
- Delivered model alerts through an Android app, an Angular vet portal and a Rasa NLP chatbot

AI & Mobile Developer Intern Jul 2024 – Aug 2024
Sagemcom Software & Technologies, R&D *Tunis, Tunisia*

- Integrated an on-device EfficientNet model (TensorFlow Lite) into a mobile app to recognise products in real time
- Cut product identification from 30 s to under 5 s and reduced manual input errors by 80%

AI Projects

GroundedDocs | *Python, FastAPI, Qdrant, LangChain, Groq, Docker, GitHub Actions*

- Built a RAG API over enterprise documents that cites every claim and refuses when evidence is missing
- Fused dense and BM25 retrieval with RRF and cross-encoder reranking: Recall@3 from 88.9% to 100%
- Verified each claim against its source: 96.3% faithfulness, 88.2% citation accuracy on 48 golden questions
- Shipped with API-key auth, rate limiting, latency and cost tracking, Docker deployment, CI and 113 tests

CourtVision | *PyTorch, YOLO11, BoT-SORT, SmolVLM2, OpenCV, TensorRT, FastAPI*

- Building a pipeline that turns NBA broadcast video into tracked players with team, number and court position
- Fine-tuned YOLO11 on 28,843 player boxes; BoT-SORT + re-ID tracking: HOTA 0.602 vs 0.393 for ByteTrack
- Mapped players to the court via keypoints and RANSAC homography: 0.695 ft median error, 398/420 teams right
- Read jersey numbers zero-shot with a vision-language model (SmolVLM2) at 99% precision

BurgerVision | *RF-DETR, YOLOv8, ONNX Runtime, OpenCV, FastAPI*

- Built an offline, real-time camera system for a fast-food client that checks each burger against its recipe
- Compared RF-DETR, YOLOv8 and D-FINE; added multi-frame confirmation, a quality gate and recipe rules
- Raised mAP50 on unseen client kitchens from 0.45 to 0.77, and reached 0.889 on the home station
- Scored 26/28 correct verdicts on held-out videos (client's own model: 21/28), with zero false approvals

AERIS | *Python, PyTorch, Audio Spectrogram Transformer, Spring Boot, Next.js*

- Designing a clinical AI prototype that joins chest X-ray, lung sounds, notes and vitals in one patient record
- Trained a heart-disease risk model at 0.89 AUROC on locked data; training an audio transformer on lung sounds
- Every AI output requires clinician approval, and the original output is kept in an audit trail

FinSight AI | *scikit-learn, XGBoost, LightGBM, statsmodels, FastAPI, Spring Boot*

- Led ML on a fintech platform: six calibrated models for fraud, anomalies, credit, forecasting and segmentation

- Fraud: XGBoost with isotonic calibration and a cost-sensitive threshold, 0.866 PR-AUC on the ULB benchmark
- Credit scorecard with exact reason codes at 0.81 AUC; cash-flow forecasts 27% better (WAPE) than baseline

RaceDeck F1 AI | *Python, FastF1, scikit-learn, MLflow, FastAPI, Docker*

- Built an MLOps platform turning Formula 1 lap data into pre-race pace, simulation and tyre-strategy briefings
- Pace model with fuel and track corrections: 1.33 s error vs 11.31 s baseline, leave-one-round-out on 11 rounds
- 4,000-run Monte-Carlo race simulator tracked in MLflow; each model ships only if it beats its baseline

Education

Engineering Degree, Big Data & Artificial Intelligence — Sep 2025 – Present
Polytech International — *Tunis, Tunisia*

Bachelor's Degree, Embedded Systems & Mobile Development — 2022 – 2025
ISET Charguia, GPA 16.78/20 — *Tunis, Tunisia*

Technical Skills

Deep Learning & CV: PyTorch, TensorFlow, YOLO, OpenCV, ONNX Runtime, TensorRT, Grad-CAM
Machine Learning: scikit-learn, XGBoost, LightGBM, CatBoost, statsmodels, SHAP, time series, NLP
LLM & RAG: LangChain, LangGraph, Qdrant, FAISS, Groq, prompt engineering, LLM evaluation
MLOps & Data: Python, SQL, pandas, NumPy, FastAPI, Docker, MLflow, GitHub Actions, pytest
Software: Java, TypeScript, C/C++, Spring Boot, Django, Next.js, Angular, Flutter, Android, ESP32, MQTT

Certifications

Samsung Innovation Campus, AI Track: multimodal capstone (EfficientNetB3, BERT, LLM), 95.5%, 4th place
AWS AI Practitioner Challenge (Udacity): LLMs, RAG and agentic AI
IT Specialist, Big Data (Certiport)

Languages

Arabic (native), French (professional), English (professional)